



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For



CompTIA Linux+ Training (XK0-006)

Course Code: C1162

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version v1

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
v1	2 Jul 2026	Initial release — CompTIA Linux+ XK0-006 V8 2-day lesson plan aligned to the 30 labs, the five exam domains and the slide deck (slide numbers cited).	Dr. Alfred Ang
v2	18 Aug 2026	Full teaching chapters from the legacy Learner Guide deck merged into the slide deck and re-ordered onto the five XK0-006 exam domains; exam-domain blueprint slide added at the start; CompTIA exam registration and practice-exam slides added at the end; labs restructured one-folder-per-lab; slide references updated to the v2 deck.	Dr. Alfred Ang
v3	18 Aug 2026	Deck condensed to the standard house slide format (525 slides): content re-authored objective-by-objective (1.1–5.5) per the XK0-006 blueprint with key screenshots retained; every lab given full step-by-step slide coverage under its exam objective; slide references updated to the v3 deck.	Dr. Alfred Ang

TABLE OF CONTENTS

Right-click and choose “Update Field”, or press F9, to build the table of contents.

Course Information

Course Title	CompTIA Linux+ Training (XK0-006)
Course Code	C1162
Certification Exam	CompTIA Linux+ XK0-006 V8
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	5 days · 7.5 instructional hours per day (37.5 hours)
Daily Timing	9:30 am – 5:30 pm (30-minute lunch break)
Mode	Instructor-led, hands-on labs on the free KillerCoda Ubuntu Playground
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- Explain core Linux concepts — the boot process, the Filesystem Hierarchy Standard, device management and virtualization — and manage storage, networking and shell operations.
- Manage files, local user and group accounts, processes and jobs, software packages, systemd services and containerized applications on a Linux server.
- Apply Linux security: authentication/authorization (sudo, PAM), firewalls, OS and account hardening, cryptography and compliance/audit procedures.
- Automate administration with Ansible, Bash and Python, apply Git version control, and use AI assistants responsibly and securely.
- Monitor a Linux system and analyze and troubleshoot hardware, storage, network, security and performance issues using the right diagnostic tools.

Exam Domain Coverage

The course maps 1:1 to the CompTIA Linux+ XK0-006 exam blueprint. Each domain, its exam weighting, the labs that cover it, and the deck slides:

Domain	Exam Weight	Labs	Deck Slides
1. System Management	23%	1–7	13–93
2. Services and User Management	20%	8–13	94–212
3. Security	18%	14–19	213–283
4. Automation, Orchestration, and Scripting	17%	20–24	284–406
5. Troubleshooting	22%	25–30	407–517

Daily Schedule

Each training day runs 9:30 am – 5:30 pm and delivers 7.5 instructional hours (450 minutes of teaching plus a 30-minute lunch break). Over five days the course delivers 37.5 instructional hours.

Day 1 — Domain 1: System Management (23%)

Time	Duration	Topic / Activity	Method	Slides
9:30–9:45	15 min	Welcome, trainer and learner introductions, ground rules	Admin	—
9:45–10:00	15 min	Course overview, learning outcomes and how the week runs	Admin	—
10:00–12:30	150 min	Domain 1 — boot & FHS, kernel modules & devices, storage/LVM. Hands-on Labs 1–3	Lecture + Hands-on Lab	13–55
12:30–1:00	30 min	Lunch break	Break	—
1:00–3:15	135 min	Domain 1 (cont.) — network configuration, shell operations. Hands-on Labs 4–5	Lecture + Hands-on Lab	56–82
3:15–5:00	105 min	Domain 1 (cont.) — backup & restore, virtualization. Hands-on Labs 6–7	Lecture + Hands-on Lab	83–90
5:00–5:30	30 min	Day 1 recap, Q&A and consolidation	Discussion	91–93

Day total: 7.5 instructional hours (9:30 am – 5:30 pm).

Day 2 — Domain 2: Services and User Management (20%)

Time	Duration	Topic / Activity	Method	Slides
9:30–9:45	15 min	Day 1 recap	Admin	—

		and Day 2 orientation		
9:45–12:30	165 min	Domain 2 — files & directories, accounts & groups, processes & jobs. Hands-on Labs 8–10	Lecture + Hands-on Lab	94–173
12:30–1:00	30 min	Lunch break	Break	—
1:00–3:15	135 min	Domain 2 (cont.) — software & package management, systemd services. Hands-on Labs 11–12	Lecture + Hands-on Lab	174–203
3:15–5:00	105 min	Domain 2 (cont.) — containers with Docker & Podman. Hands-on Lab 13	Lecture + Hands-on Lab	204–209
5:00–5:30	30 min	Day 2 recap, Q&A and consolidation	Discussion	210–212

Day total: 7.5 instructional hours (9:30 am – 5:30 pm).

Day 3 — Domain 3: Security (18%)

Time	Duration	Topic / Activity	Method	Slides
9:30–9:45	15 min	Day 2 recap and Day 3 orientation	Admin	—
9:45–12:30	165 min	Domain 3 — AAA (sudo, PAM, Polkit), firewalls. Hands-on Labs 14–15	Lecture + Hands-on Lab	213–263
12:30–1:00	30 min	Lunch break	Break	—
1:00–3:15	135 min	Domain 3 (cont.) — OS hardening, account hardening. Hands-on Labs	Lecture + Hands-on Lab	264–270

		16–17		
3:15–5:00	105 min	Domain 3 (cont.) — cryptography, compliance & auditing. Hands-on Labs 18–19	Lecture + Hands-on Lab	271–278
5:00–5:30	30 min	Day 3 recap, Q&A and consolidation	Discussion	279–283

Day total: 7.5 instructional hours (9:30 am – 5:30 pm).

Day 4 — Domain 4: Automation, Orchestration, and Scripting (17%)

Time	Duration	Topic / Activity	Method	Slides
9:30–9:45	15 min	Day 3 recap and Day 4 orientation	Admin	—
9:45–12:30	165 min	Domain 4 — Infrastructure as Code with Ansible. Hands-on Lab 20	Lecture + Hands-on Lab	284–360
12:30–1:00	30 min	Lunch break	Break	—
1:00–3:15	135 min	Domain 4 (cont.) — Bash scripting, Python for sysadmin. Hands-on Labs 21–22	Lecture + Hands-on Lab	361–393
3:15–5:00	105 min	Domain 4 (cont.) — Git version control, responsible AI use. Hands-on Labs 23–24	Lecture + Hands-on Lab	394–401
5:00–5:30	30 min	Day 4 recap, Q&A and consolidation	Discussion	402–406

Day total: 7.5 instructional hours (9:30 am – 5:30 pm).

Day 5 — Domain 5: Troubleshooting (22%)

Time	Duration	Topic / Activity	Method	Slides
9:30–9:45	15 min	Day 4 recap and Day 5 orientation	Admin	—
9:45–12:30	165 min	Domain 5 — monitoring	Lecture + Hands-on Lab	407–471

		concepts & tools, hardware/storage/OS triage. Hands-on Labs 25–26		
12:30–1:00	30 min	Lunch break	Break	—
1:00–3:15	135 min	Domain 5 (cont.) — network and security troubleshooting . Hands-on Labs 27–28	Lecture + Hands-on Lab	472–495
3:15–4:45	90 min	Domain 5 (cont.) — performance troubleshooting . Hands-on Lab 29	Lecture + Hands-on Lab	496–502
4:45–5:30	45 min	Capstone — end-to-end server build and triage. Hands-on Lab 30; course wrap-up	Lecture + Hands-on Lab	503–517

Day total: 7.5 instructional hours (9:30 am – 5:30 pm).

Topic-by-Topic Breakdown

Each lab below cites the deck slide of its overview so the trainer can jump straight to it.

Domain 1 — System Management (23%) · Slides 13–93

Lab 1 — Boot Process & Filesystem Hierarchy [Obj 1.1] · Slide 42

Inspect a running Linux system's boot configuration, kernel command line, initramfs, and the standard Filesystem Hierarchy to understand how GRUB hands off to the kernel and where FHS directories live.

Lab 2 — Kernel Modules & Device Management [Obj 1.2] · Slide 46

List, inspect, load, and unload kernel modules and use the standard hardware-enumeration tools to understand modprobe vs insmod, module dependencies, and reading dmesg.

Lab 3 — Storage with LVM, Partitions & Filesystems [Obj 1.3] · Slide 59

Build a complete LVM stack on loopback-backed disks, format with ext4/xfs/btrfs, mount with hardening options, persist in fstab, and grow a filesystem online.

Lab 4 — Network Configuration [Obj 1.4] · Slide 82

Inspect interfaces, addresses, and routes with `ip`; walk the name-resolution stack; validate a `netplan` YAML; and use the standard diagnostic toolbox to read a Linux network stack top-to-bottom.

Lab 5 — Shell Operations & Text Processing [Obj 1.5] · Slide 86

Work with shell environment variables, login vs interactive startup files, every common redirection form, and the classic text-processing pipeline to assemble non-trivial one-liners.

Lab 6 — Backup & Restore [Obj 1.6] · Slide 90

Create `tar/cpio/dd` archives with `gzip/bzip2/xz` compression, image and recover disks with `dd` and `ddrescue`, mirror trees with `rsync`, and verify integrity with `sha256sum`.

Lab 7 — Virtualization with QEMU/KVM & libvirt [Obj 1.7] · Slide 94

Install `QEMU` and `libvirt`, manipulate virtual disk images with `qemu-img`, list `libvirt` domains/networks with `virsh`, and boot a tiny guest under `QEMU` to discuss hypervisors, image formats, snapshots, and network modes.

Domain 2 — Services and User Management (20%) · Slides 94–212

Lab 8 — File & Directory Management [Obj 2.1] · Slide 123

Practice core file and directory operations: navigating, creating, copying, searching, linking, and inspecting special device files with standard Linux utilities.

Lab 9 — Local Account & Group Management [Obj 2.2] · Slide 138

Create and manage local users and groups, set passwords and aging policies, change shells, and distinguish user, system, and service accounts.

Lab 10 — Processes, Jobs & Scheduling [Obj 2.3] · Slide 159

Inspect processes, manipulate shell jobs, adjust priorities, send signals, and schedule tasks with `at`, `cron`, and `anacron`.

Lab 11 — Software & Package Management [Obj 2.4] · Slide 192

Manage Debian packages with `apt/dpkg`, add a signed third-party repo, run `dnf/rpm` in a container, and use language and sandboxed package managers.

Lab 12 — Service Management with systemd [Obj 2.5] · Slide 196

Inspect and control `systemd` units, author custom service/timer/mount units, analyze boot performance, and configure host/time/DNS via `systemd` helpers.

Lab 13 — Containers with Docker & Podman [Obj 2.6] · Slide 213

Install `Docker` and `Podman`, pull/run images, manage volumes and networks, build an image from a `Dockerfile`, and contrast rootful vs rootless engines.

Domain 3 — Security (18%) • Slides 213–283

Lab 14 — AAA: sudo, PAM, and Polkit [Obj 3.1] • Slide 238

Configure sudo authorization, inspect PAM authentication stacks, and enable auditd accounting to record security events. Finish with a conceptual look at centralized identity via LDAP, Kerberos, and SSSD.

Lab 15 — Firewalls: ufw, firewalld, iptables, nftables, ipset [Obj 3.2] • Slide 253

Configure firewalls at every layer of the Linux netfilter stack, from the ufw wrapper down to iptables and nftables, and understand stateful inspection. Explore zones, rich rules, ipset, and SNAT/DNAT.

Lab 16 — OS Hardening: Permissions, SELinux, SSH, Fail2ban [Obj 3.3] • Slide 272

Apply layered OS hardening from file permissions and immutability through SELinux labeling, then harden OpenSSH and add brute-force protection. Inventory SUID binaries and legacy cleartext services.

Lab 17 — Account Hardening [Obj 3.4] • Slide 276

Harden user accounts through password quality and aging policies, brute-force lockouts, and restricted or non-interactive shells. Explore TOTP MFA and Have I Been Pwned k-anonymity lookups.

Lab 18 — Cryptography: GPG, LUKS, OpenSSL, WireGuard [Obj 3.5] • Slide 280

Practice core Linux cryptography: GPG key generation and signing, LUKS2 disk encryption, OpenSSL certificates, hashing and HMAC, and WireGuard keys. Learn to identify weak TLS versions and ciphers.

Lab 19 — Compliance, Auditing, and File Integrity [Obj 3.6] • Slide 284

Establish file-integrity monitoring and run compliance and audit tooling to detect tampering, rootkits, and misconfigurations. Cover secure deletion, login banners, and OpenSCAP/CIS benchmarks.

Domain 4 — Automation, Orchestration, and Scripting (17%) • Slides 284–406

Lab 20 — Infrastructure as Code with Ansible [Obj 4.1] • Slide 314

Learn Infrastructure as Code using Ansible as the primary agentless configuration management tool, and recognize the wider IaC ecosystem (Puppet, OpenTofu, cloud-init, Kubernetes ConfigMaps). Build an idempotent playbook that deploys and configures nginx.

Lab 21 — Bash Scripting [Obj 4.2] • Slide 369

Practice core Bash scripting constructs (variables, conditionals, loops, functions, option parsing, regex, file tests) and assemble a defensive backup script, then audit it with shellcheck.

Lab 22 — Python for System Administration [Obj 4.3] • Slide 373

Use Python for common Linux admin tasks: create a virtual environment, exercise core data types, walk the filesystem, parse JSON, call a REST API with requests, and enforce PEP 8 with black and flake8.

Lab 23 — Git Version Control [Obj 4.4] · Slide 403

Work through Git end to end: init, commit, branch, merge, resolve conflicts, stash, tag, rewrite history with reset and rebase, and push/pull against a local bare remote.

Lab 24 — Responsible AI Use for Linux Administrators [Obj 4.5] · Slide 407

Adopt a verify-before-paste workflow for AI-generated Linux commands: redact secrets before sending prompts to hosted models, lint and sandbox suggestions, weigh local vs hosted models, and draft a corporate AI usage policy.

Domain 5 — Troubleshooting (22%) · Slides 407–517

Lab 25 — Monitoring Concepts and Tools [Obj 5.1] · Slide 416

Learn the monitoring vocabulary (SLA/SLO/SLI) and stand up a working monitoring stack to produce metrics, alerts, webhooks, health checks, and aggregated logs.

Lab 26 — Troubleshooting Hardware, Storage, and OS Issues [Obj 5.2] · Slide 459

Deliberately inject common OS and storage failures then diagnose and repair each, keeping every breakage contained to loopback devices, throwaway units, or subshells.

Lab 27 — Troubleshooting Network Connectivity [Obj 5.3] · Slide 480

Break and repair the common network faults (DNS, routing, firewall, MTU, IP, link, dual-stack) while practicing the standard L3/L4 triage toolkit.

Lab 28 — Troubleshooting Security Issues [Obj 5.4] · Slide 492

Diagnose and repair security faults spanning SELinux contexts, POSIX ACLs, certificate trust, account lockout, broken repos, and weak TLS, using a Rocky container for the SELinux portion.

Lab 29 — Troubleshooting Performance Issues [Obj 5.5] · Slide 508

Stress CPU, memory, disk I/O, and network in turn, then walk the universal performance loop: baseline, spike, identify the bottleneck with the right counter, and resolve it.

Lab 30 — Capstone: End-to-End Server Build and Triage [Obj Capstone] · Slide 511

Integrate Objectives 1-5 into one coherent build: a hardened admin account, LVM storage, nginx with a systemd override, a locked firewall, a health-check timer, and git-versioned config, then inject a fault and triage it end-to-end.

Resources Required

- A laptop with a modern web browser and reliable internet access.

- Free Killercoda Ubuntu Playground — <https://killercoda.com/playgrounds/scenario/ubuntu> (no local install required).
- Course slide deck and Learner Guide (downloaded from the LMS).
- Tertiary Infotech LMS account — <https://lms-tms.tertiaryinfotech.com/> (courseware download and assessment).
- CompTIA Linux+ XK0-006 V8 Exam Objectives (reference blueprint).